DeFi–CeFi Flow Internalization

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Abstract

A market maker filling 1inch Fusion intents faces a per-fill choice: **hedge** the acquired inventory immediately on a CEX, or **warehouse** it and wait for offsetting flow. We study this decision for the WETH/USDT pair using 1,017 Fusion fills (5–7 May 2025) matched against 6,600 minute-stamped Binance ETH/USDT order-book snapshots. 3 results:

- Actual hedge slippage on fully executable Binance snapshot fills is near-zero and approximately affine in size (0.021 + 0.0056·s bps); depth-limited fills and stress regimes remain the impact-risk caveats.
- Flow toxicity is driven by **size**, not short-window price correlation — fills >10 ETH mark out –10.7 bps against the warehouse over 60 minutes while 0.3–1.29 ETH flow marks out **+4.7 bps**.
- In an 858-fill policy replay, a size-conditional selective engine loses $704 vs –$1,247 always-hedge and –$5,426 always-warehouse, with an $8.4k oracle bound — flow capture on this sample is near break-even, and headline backtest PnL (+$90k) is directional ETH exposure, not strategy alpha.

1 · Strategy and decision framework

1inch Fusion is an intent-based Dutch-auction protocol: resolvers compete to fill user swap orders, typically printing inside the CEX spread. A resolver internalizing WETH/USDT flow accumulates inventory whose risk must be either offset on Binance (paying taker fee + slippage, locking the captured edge) or warehoused (keeping the edge, wearing the mark-to-market drift until natural offset arrives). Per fill of size s ETH at edge e bps vs the Binance mid, the engine compares the 2 expected PnLs and applies a size gate.

The punchline up front: the warehouse decision alone is not a large alpha source in this sample — its role is loss avoidance. The monetizable edge sits upstream, in intent pricing, fill selection, and avoiding toxic large flow.

Data

Dataset	Source	Size	Notes
1inch Fusion fills, WETH/USDT	Dune (saved query 4966942)	1,017 fills	block-time, size, px, direction, tx
Binance ETH/USDT spot book	REST snapshots, 1-min cadence	6,600 files	OHLCV + top-of-book bids/asks
Window	5 May 08:04 – 7 May 09:08 UTC	~49 h	ETH rallied ≈ +9% over the window
Fill sizes	median 0.31 ETH · max 640 ETH	—	heavily fat-tailed

2 · Model specification

With N = s·mid the per-fill notional, fee the taker fee (bps), slip(s) the calibrated slippage curve (bps), tox(s) the expected adverse 60-minute markout of warehoused flow (bps, defined below), and δ the toxicity fraction realized during the hedge delay:

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tox(s) = –E[ signed warehouse markout | size = s ]   (sign: + = adverse cost)
slip(s) = (0.0206 + 0.00562·s) × stress    δ = max(tox(s),0) × delay/3600
PnL_hedge     = N · (e – fee – slip(s) – δ) / 1e4
PnL_warehouse = N · (e – tox(s)) / 1e4
decision: HEDGE if PnL_hedge > PnL_warehouse, or (s > threshold and tox(s) > 0); else
WAREHOUSE
```

Note that the captured edge e enters both branches identically and cancels from the comparison: the decision is driven entirely by hedge costs vs size-conditional toxicity. With calibrated parameters the effective hedge boundary sits at ≈2.9 ETH — where tox(s) crosses zero — rather than at the nominal 1.29 ETH optimizer threshold.

Calibrated parameters

Parameter	Value	Source
Binance taker fee	6.66 bps	backtest optimizer (optimized_params)
Slippage curve	0.021 + 0.0056·s bps	affine fit, 316 actual VWAP fills
Hedge delay	33 s	optimizer; consistent with 1–2 min DEX→CEX lead
Size threshold	1.29 ETH	optimizer; superseded in effect by tox(s) zero-crossing
Toxicity curve tox(s)	piecewise-linear in log s	60-min signed markouts (Section 3)

Slippage calibration

Measured against the 316 fills that completed at order-book VWAP, actual slippage is 0.01–2 bps even >100 ETH — the book absorbs essentially all observed flow, and an affine fit in size (0.021 + 0.0056·s bps) describes it well. On fully executable snapshot fills, hedging appears cheap across the observed size range; excluded depth-limited cases and stress regimes remain the main impact-risk caveats. 129/445 candidate fills could not be fully filled at snapshot top-of-book depth and are excluded from the fit (real depth is deeper than the snapshots).

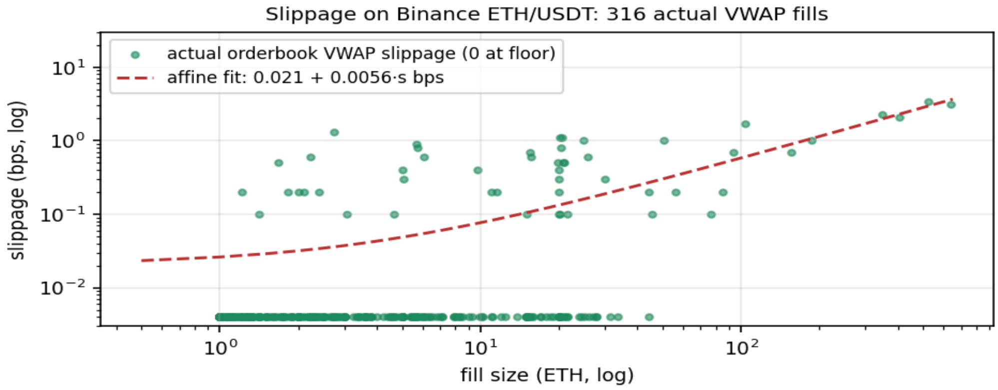

Fig1. Actual order-book VWAP slippage by fill size with the affine fit.

3 · Flow toxicity: size beats correlation

For every fill we compute the **signed 60-minute markout**: the Binance mid move after the fill, signed by the inventory direction the warehouse would hold (+ = warehoused position profits). This is the realized benefit or cost of not hedging. We bucket markouts by the candidate toxicity signals the original pipeline computed: ±2.5-min price correlation, ±2.5-min volume correlation, and fill size. The engine cost tox(s) of Section 2 is the negative of this bucket mean. Size-bucket rows use the 858 fills with valid 60-minute forward bars; correlation rows are computed on the subset of fills with available correlation features from the full 1,017-fill sample.

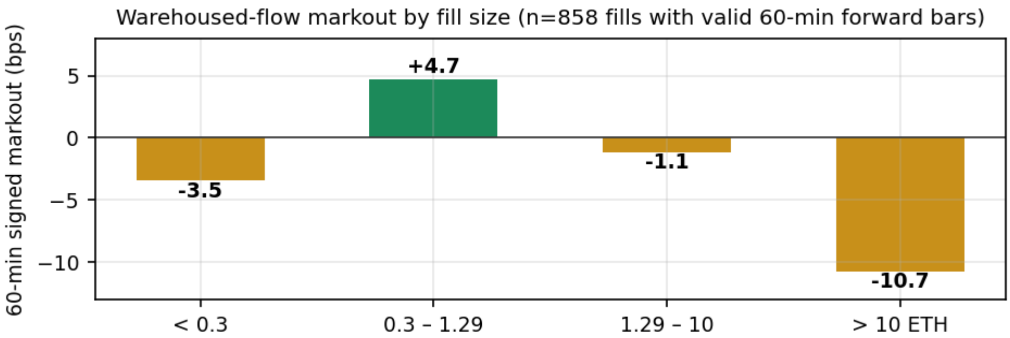

Fig2. Mean signed 60-min markout of warehoused flow by fill size.

Signal / bucket	n	15 m	60 m	95% CI (60 m)	Verdict
|price_corr| < 0.3	123	+2.3	+0.7	—	non-monotone → rejected
|price_corr| 0.3–0.7	156	+0.6	+5.9	—	—
|price_corr| ≥ 0.7	219	+1.7	–1.9	—	—
volume_corr ≥ 0.4 (vs < 0.4)	94	–0.1	–2.6	—	weak secondary; parked
size < 0.3 ETH	414	–1.3	–3.5	[–8.5, +1.8]	mildly adverse
size 0.3–1.29 ETH	223	+1.7	+4.7	[–2.3, +12.0]	benign — warehouse pays
size 1.29–10 ETH	157	+2.1	–1.2	[–9.4, +6.9]	marginal
size > 10 ETH	64	–0.3	–10.7	[–23.8, +2.8]	toxic — hedge fast

The contemporaneous ±2.5-minute price correlation — computed but never consumed by the original Step2b code — is **not predictive**: markouts across its buckets are non-monotone at every horizon. Size is the clearest candidate toxicity axis in this sample: bucket means are economically coherent — mid-size flow is benign, larger flow is increasingly adverse, and the >10 ETH bucket carries the main warehouse risk (consistent with the Granger result that 1inch leads Binance by 1–2 minutes) — though individual 60-minute bucket CIs overlap zero. The engine therefore prices warehouse risk with a piecewise-linear toxicity curve in log-size, with a manual regime multiplier; rolling re-calibration is the planned next upgrade. Each bucket's 60-minute CI individually overlaps zero (bootstrap, size rows above) — the toxicity story rests on the cross-bucket pattern and the bucketed policy replay of Section 4, not on any single bucket's significance.

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IMPLEMENTATION RULE OF THUMB  Warehouse mid-size flow (0.3–1.29 ETH); hedge large flow (the curve flips at ≈2.9 ETH, hard gate >10 ETH); ignore short-window price correlation until a stronger signal is validated.
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4 · Backtest and policy replay

The inventory-aware backtest (100 ETH + 100k USDT start) ends +$90,263 on $283k (+31.9% over ~2 days). Attribution shows why this headline flatters: the window coincided with an ETH rally, and the directional inventory mark dominates. Spread/decision PnL is an order of magnitude smaller.

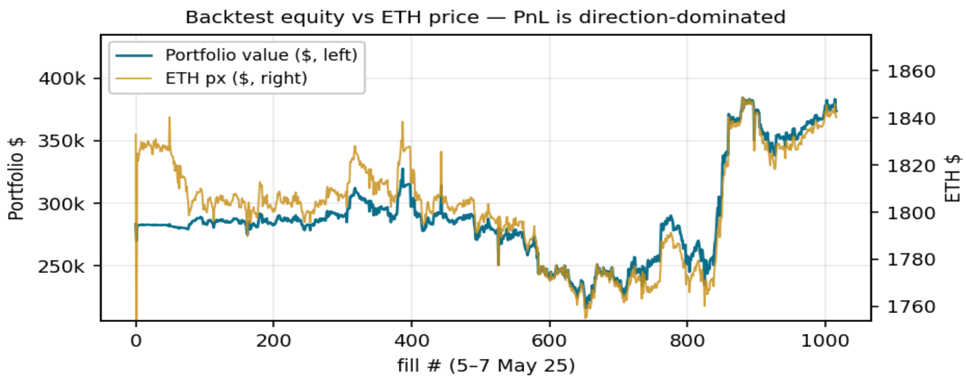

Fig3. Portfolio value tracks the ETH price almost 1-for-1.

To isolate decision quality we replay 858 fills (those with valid 60-minute forward bars) and credit each policy with realized PnL: hedge = edge – fee – slip(s); warehouse = edge + realized markout, vs the Binance mid at the fill minute. 95% intervals are from a moving-block bootstrap (blocks of 50 consecutive fills, 2,000 resamples).

Policy	PnL ($)	95% CI	Comment
Oracle (per-fill best)	+8,391	[+3,812, +15,440]	hindsight upper bound — see note
Backtest recommendation	–531	—	look-ahead, not implementable
Selective engine	–704	[–2,357, +1,150]	ex-ante information only; 133 hedges
Always hedge	–1,247	[–2,898, +726]	fee + slip on every fill
Always warehouse	–5,426	[–19,951, +5,097]	wears the >10 ETH toxicity

The strongest single result: while absolute totals are noisy, the **selective-minus-always-hedge gap is robust under the bootstrap — 95% CI +$92 to +$1,044, P(selective > always-hedge) = 0.99**.

By size bucket the engine wins for the stated reason — it warehouses the benign band and hedges every >10 ETH fill:

Size bucket	n	Hedges	Selective $	Always hedge $	Always wh $
< 0.3 ETH	414	0	–3	–37	–3
0.3–1.29 ETH	223	0	+224	–122	+224
1.29–10 ETH	157	69	–140	–304	+19
> 10 ETH	64	64	–785	–785	–5,666

3 readings:

- The selective engine recovers most of the gap between the naive policies and sits within $173 of the look-ahead benchmark using only ex-ante information.
- All implementable policies are near break-even: on this sample, edge capture roughly pays for hedging costs and toxicity — the economics of the strategy live in fill selection and auction pricing, not in the warehouse decision alone.
- The $8.4k oracle bound suggests headroom for a better toxicity signal (regime-aware curves, order-flow imbalance, cross-venue lead-lag). It is not expected attainable PnL — it is an upper bound on per-fill classification value under perfect hindsight.

5 · Extensions tested — 2 negative results

Spread stat-arb (rejected). An earlier notebook proposed z-score mean reversion on the 1inch–Binance spread, exploiting the measured 1-minute DEX→CEX lead, and reported Sharpe 60–122 with 89–95% win rates. It does not replicate: on the same 2,020 joint minutes with next-minute execution and the calibrated 13.3 bps round-trip cost, every parameter combination loses 8–27% (win rates 13–20%); the best loses 5.6% even at zero cost. The notebook's total-return column was NaN throughout — its execution embedded look-ahead. The lead is real but too small vs spread noise and costs to trade as minute-bar mean reversion.

Per-fill ML toxicity (not yet viable). A walk-forward logistic regression predicting each fill's markout *sign* from 7 ex-ante features (log size, side, edge, 15-min flow imbalance and count, 30-min momentum and realized vol) achieves AUC 0.533 ± 0.32 vs 0.525 for size alone — per-fill toxicity sign is essentially unpredictable on 1,250 fills from 1 regime. Only the size-conditional *mean* (the toxicity curve of Section 3) carries signal. An LSTM(16) sequence classifier (ported from a prior project that trained on ~800 days of candles) is implemented but gated behind ≥60 days of accumulated live snapshots to avoid regime memorization; the production updater archives its snapshots automatically. A **risk gate** (inventory cap forcing hedges on projected breaches) was also added to the engine from the earlier risk-manager design.

6 · Limitations

Sample. 2 days, 1 pair, 1 venue, a rallying regime: every calibrated number is in-sample. The toxicity curve and slippage fit should be re-estimated on rolling windows before live reliance. **Slippage.** The affine fit is measured on a deep book; depth can vanish in stress (129/445 large fills already exceeded snapshot top-of-book depth). The stress multiplier is a manual, not modeled, control. **Toxicity.** A flat 60-minute horizon proxies the warehouse holding period; true natural-offset arrival times are flow-dependent. **Costs excluded.** Funding, borrow, venue-transfer costs and capital usage are out of scope; inventory is assumed pre-positioned on both venues. **Microstructure.** Live decisions use snapshot mid/VWAP, not queue simulation; the 33 s hedge delay is taken as fixed. **What changes live?** Stale Dune results, realized hedge latency vs the fixed 33 s, Binance depth decay and fee-tier changes, venue outages, binding inventory caps — each shifts the hedge/warehouse boundary; all argue for rolling re-calibration.

7 · Production architecture (summary)

The research stack compiles to a single self-contained HTML dashboard (offline, no dependencies) carrying the embedded backtest, decision calculator, and live blocks, refreshed every 5 minutes by a macOS launchd job (Binance public REST; Dune saved query with stale-guard; live → last-known fallback; guarded atomic embeds). The engine exists in JS and Python with enforced parity.

8 · Next steps

Item	What	Status
LSTM toxicity model	train once ≥60 days of live snapshots accumulate (script + data gate in place)	gated
Rolling calibration	toxicity + slippage curves re-fit on trailing windows in the updater	planned
Multi-pair extension	recovered WAVAX/USDT Dune query generalizes to other pairs	candidate
Queue/impact model	replace snapshot VWAP with depth-consuming execution sim	candidate